

Customer, Product and Profitability Performance Analysis in Supply Chain Operations

Commercial Intelligence Project for APL Logistics (KWE Group)

Total Revenue \$36.78M	Total Profit \$3.97M	Profit Margin 10.78%	Orders Analyzed 180,519
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This report presents a practical profitability analysis built from a cleaned supply chain dataset and an interactive Streamlit dashboard. The writing is intentionally kept clear and business-focused so the findings can be understood by both technical reviewers and decision makers.

Abstract

A logistics business can show strong sales while still losing profit through discounts, weak customer mix, and high-cost regional operations. This project studies 180,519 transactions from the APL Logistics supply chain dataset to understand where value is actually created. Instead of looking only at sales volume, the analysis connects revenue, profit, customer value, product category performance, discount behavior, and regional profitability.

The dashboard results show total revenue of \$36,784,734.31 and total profit of \$3,966,902.97, giving an overall profit margin of 10.78%. Customer analysis shows that profit is not spread evenly across the customer base. Product analysis shows that some categories produce strong margins while others, especially Strength Training, contribute very little margin. Discount simulation suggests that applying a 15% discount cap could improve projected profit to about \$4.40M and raise margin to 11.95%.

The main conclusion is simple: APL Logistics should not judge commercial success only by sales. Profit-aware pricing, customer segmentation, discount control, and regional monitoring can help the business protect margin while still supporting growth.

1. Introduction

Supply chain companies often focus on delivery speed, order volume, and revenue growth. These metrics are important, but they do not fully explain whether the business is financially healthy. A customer can place many orders and still be unprofitable. A product can generate high revenue and still have a weak margin. A market can look attractive from a sales perspective but may not generate enough profit after discounts and operational costs are considered.

For a global logistics provider such as APL Logistics, this creates a real management challenge. Leadership needs to know which customers, products, categories, and regions are worth prioritizing. The goal of this project is to convert detailed order data into commercial intelligence that supports better pricing, customer targeting, and margin decisions.

2. Problem Statement

The organization has access to detailed order and sales data, but the data needs to be translated into profitability insight. The main business problem is that revenue-focused analysis can hide margin leakage. Discounts, low-margin product categories, and loss-making customers may reduce profit even when sales appear healthy.

This project answers four practical questions:

- Which customer segments and customer groups contribute the most value?
- Which products and categories are profitable, and which ones need attention?
- How much can discount control improve profitability?
- Which markets and regions show the strongest commercial performance?

3. Dataset and Preparation

The cleaned dataset contains 180,519 order records and 19 commercial fields. The key columns used in the analysis include customer details, customer segment, product category, product name, market, order region, sales, order profit per order, discount value, discount rate, profit margin, delivery status, late delivery indicator, and shipping mode.

The preparation stage focused on keeping the dashboard stable and easy to use. A full customer name field was created where needed, missing or invalid values were handled, and margin fields were calculated using safe division logic to avoid errors when filters returned a small or zero-sales subset. This step was important because the

dashboard allows users to filter by segment, market, region, category, product, and discount scenario.

4. Methodology

The analysis followed a structured workflow: data cleaning, KPI calculation, customer contribution analysis, product and category margin analysis, discount impact simulation, and market-level diagnostics. The dashboard was developed in Streamlit using pandas for aggregation and Plotly for interactive charts.

Analytical Area	Method Used	Decision Supported
Revenue and Profit Overview	Total sales, total profit, margin calculation	Understand overall commercial health
Customer Value	Segment contribution, top and bottom customers, value tiers, Pareto curve	Prioritize retention and account review
Product and Category Performance	Product ranking, category margin, category-region heatmap	Improve pricing and product strategy
Discount Impact	Discount rate vs margin chart and 15% cap simulation	Control margin erosion from excessive discounts
Market and Region Analysis	Market margin, regional profit, country map	Support regional strategy and expansion decisions

The what-if discount simulation assumes that discount amounts above the selected cap are recovered as saved profit. This is a directional business model, not a complete demand elasticity model. In real operations, the final policy should also consider customer response, contract terms, and competitive pressure.

5. Revenue and Profit Overview

The first dashboard view summarizes the overall financial position. The full dataset produced \$36.78M in revenue and \$3.97M in profit, which gives a profit margin of 10.78%. This confirms that the company is profitable overall, but the margin is narrow enough that pricing and discount decisions can have a major effect.

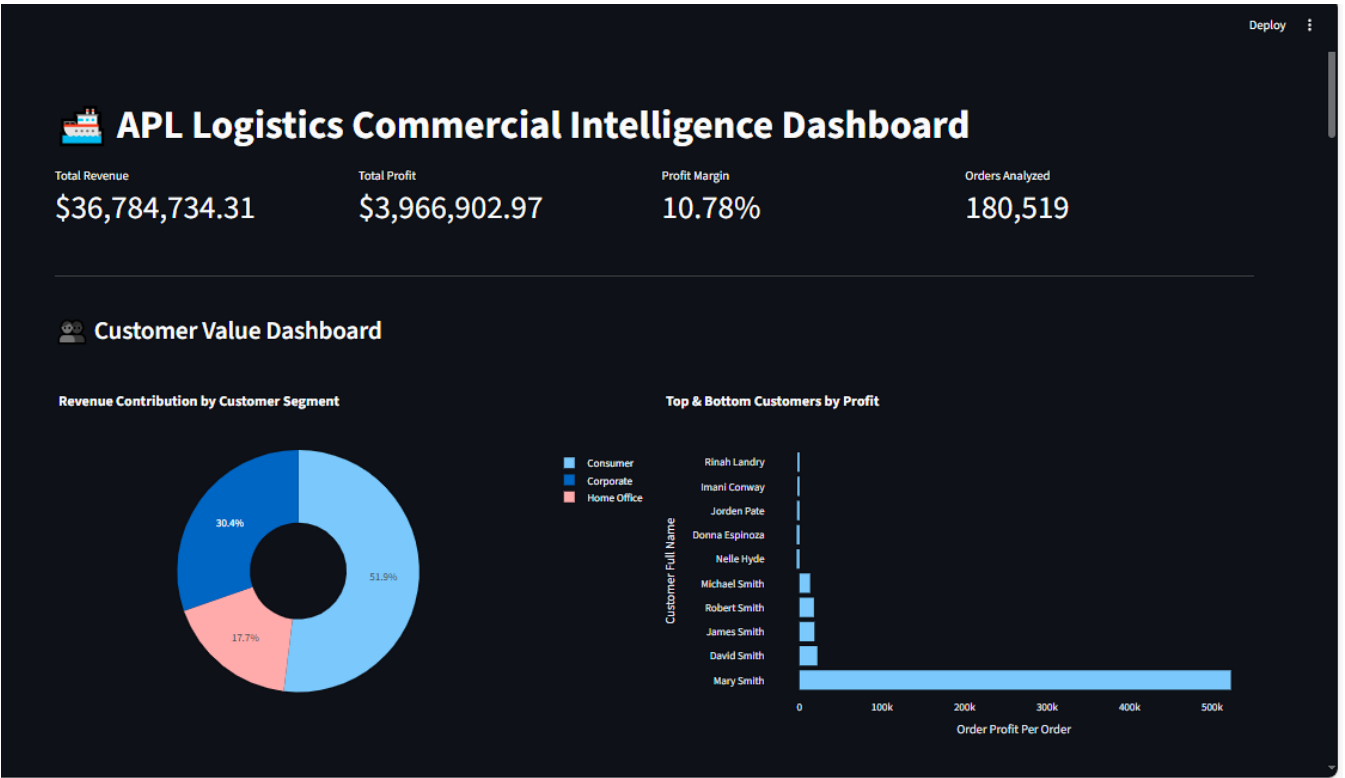


Figure 1. Executive dashboard view showing total revenue, total profit, profit margin, orders analyzed, and customer segment contribution.

6. Customer Value Analysis

The customer dashboard shows that Consumer customers account for the largest revenue contribution at about 51.9%, followed by Corporate customers at 30.4% and Home Office customers at 17.7%. This does not mean Consumer customers should automatically receive the highest priority, but it does show where the largest revenue base currently sits.

The dashboard also separates customers into Bronze, Silver, Gold, and Platinum value tiers. This gives the business a simple way to identify customers that deserve retention focus and customers that may require pricing review. A customer can be high volume but low value if discounts or fulfillment costs reduce profit.

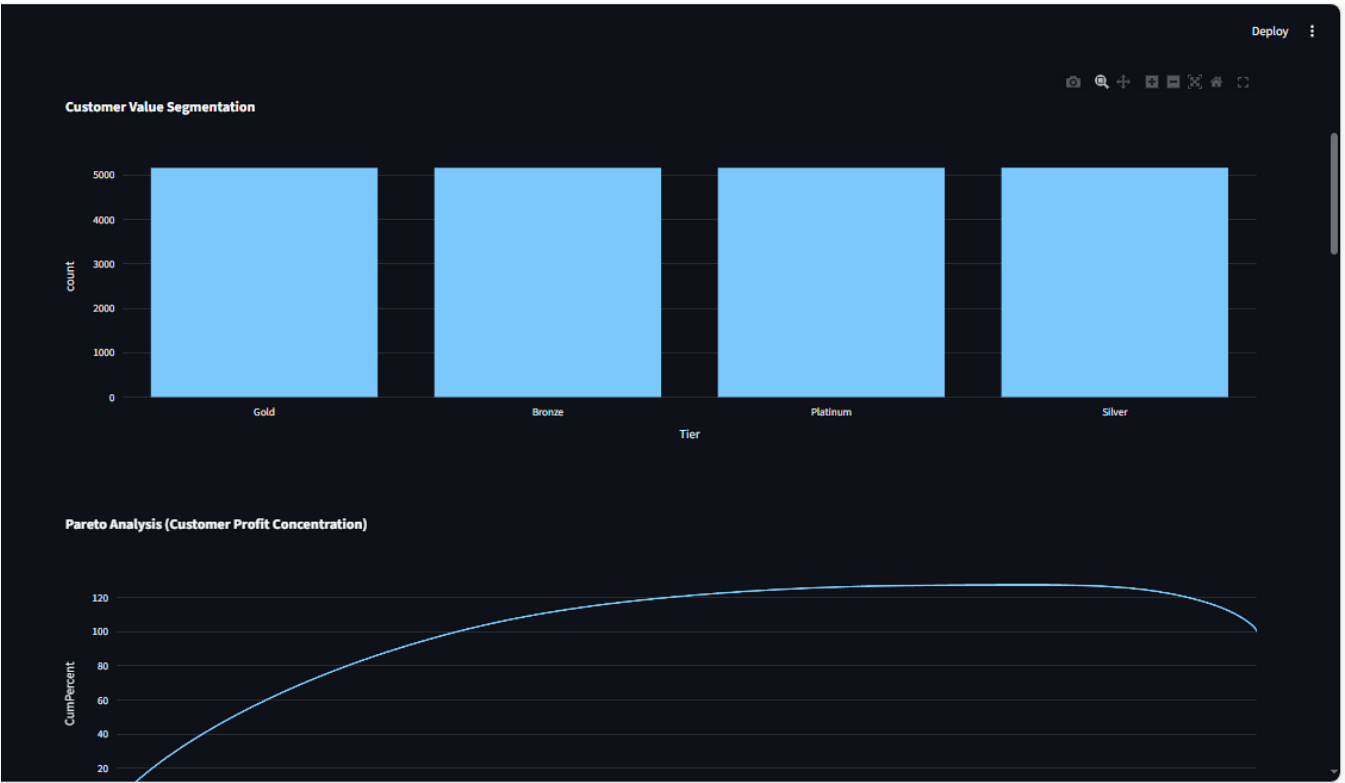


Figure 2. Customer value segmentation showing the distribution of customers across Bronze, Silver, Gold, and Platinum tiers.

The Pareto curve adds an important insight. The curve rises above 100% before settling back because profitable customers generate more than the final net profit, while loss-making customers reduce the total. This is not just a chart behavior - it is a commercial warning. It means that some customer groups are actively pulling down the profit created by stronger accounts.

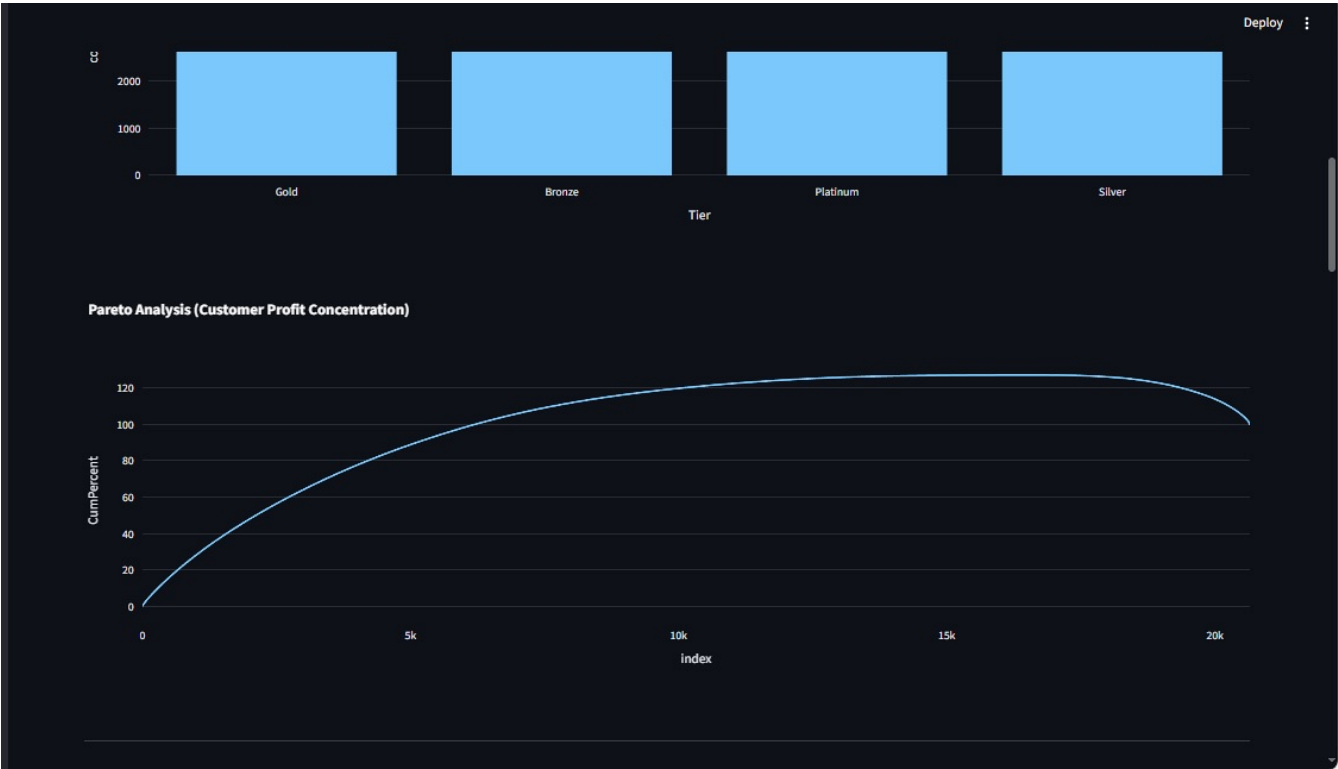


Figure 3. Pareto analysis showing customer profit concentration and the impact of lower-value customers on final net profit.

7. Product and Category Profitability

Product analysis shows that profit is strongly concentrated in a small set of products. The leading product by total profit is Field & Stream Sportsman 16 Gun Fire Safe, followed by Perfect Fitness Perfect Rip Deck and Diamondback Women's Serene Classic Comfort Bike. These products are important because they generate meaningful profit, not just sales.

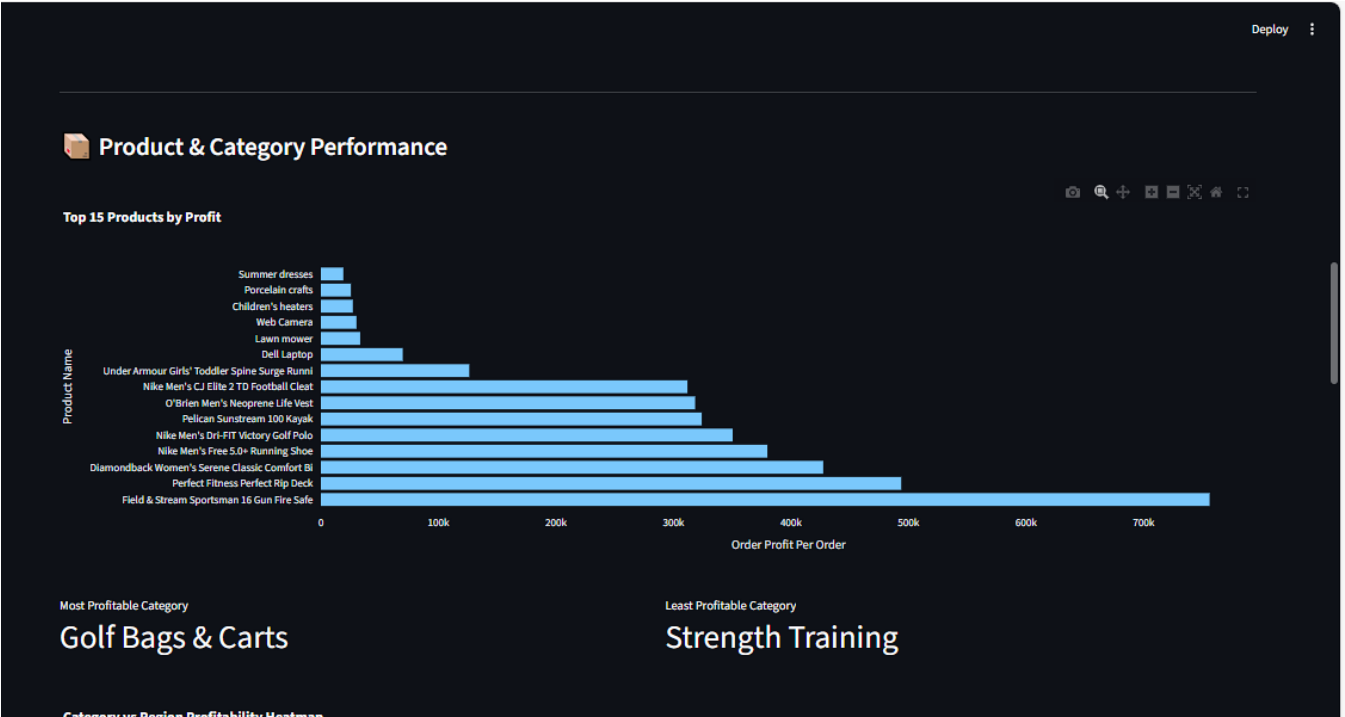


Figure 4. Product and category performance dashboard showing the top 15 products by profit and category margin indicators.

Category analysis gives a more strategic view. Fishing is the highest absolute profit contributor, but Golf Bags & Carts shows the strongest category margin at 17.46%. Strength Training is the weakest category, with a margin of only 0.61%. This indicates that Strength Training may need pricing review, freight cost review, or a change in promotional strategy.

Category Insight	Result	Interpretation
Highest total profit category	Fishing - \$756,220.76	Large category-level profit engine
Highest margin category	Golf Bags & Carts - 17.46%	Strongest return per sales dollar
Lowest margin category	Strength Training - 0.61%	Requires pricing or cost review
Major profitable products	Gun safe, fitness deck, comfort bike, running shoe	Useful for sales and inventory prioritization

8. Category and Region Heatmap

The heatmap compares product categories across order regions. This is useful because a category may perform well globally but underperform in a specific market or shipping lane. For example, heavy or bulky categories can lose margin in regions where freight and fulfillment costs are higher.

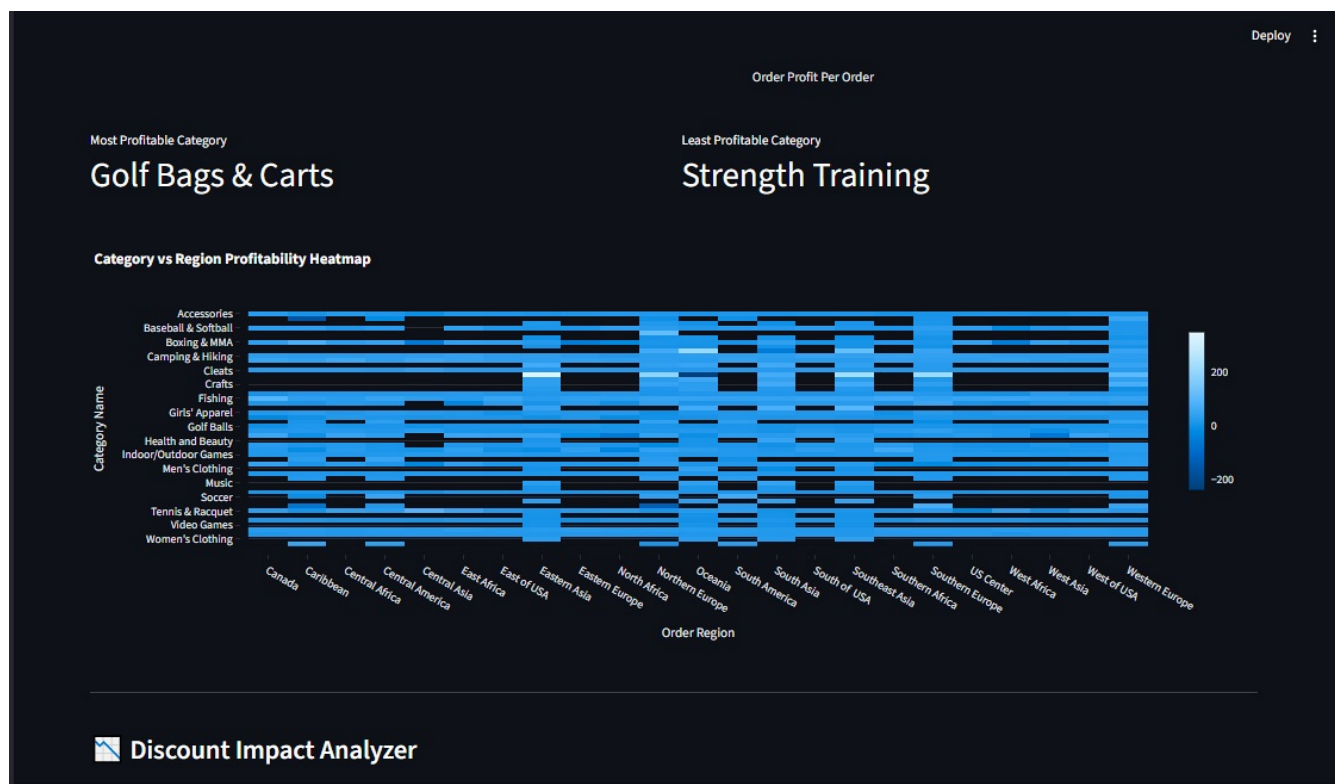


Figure 5. Category vs region profitability heatmap used to locate product-region combinations with stronger and weaker average profit.

This view is valuable for regional managers because it moves the analysis beyond a single global average. Instead of asking whether a product category is good or bad overall, the dashboard helps ask where the category is profitable and where it needs corrective action.

9. Discount Impact Diagnostics

Discounting is one of the clearest sources of margin pressure in the dataset. The discount analyzer compares order item discount rate with profit margin. The chart shows that higher discount rates are often associated with lower and sometimes negative margins. This supports the business concern that discounts can silently erode profitability.

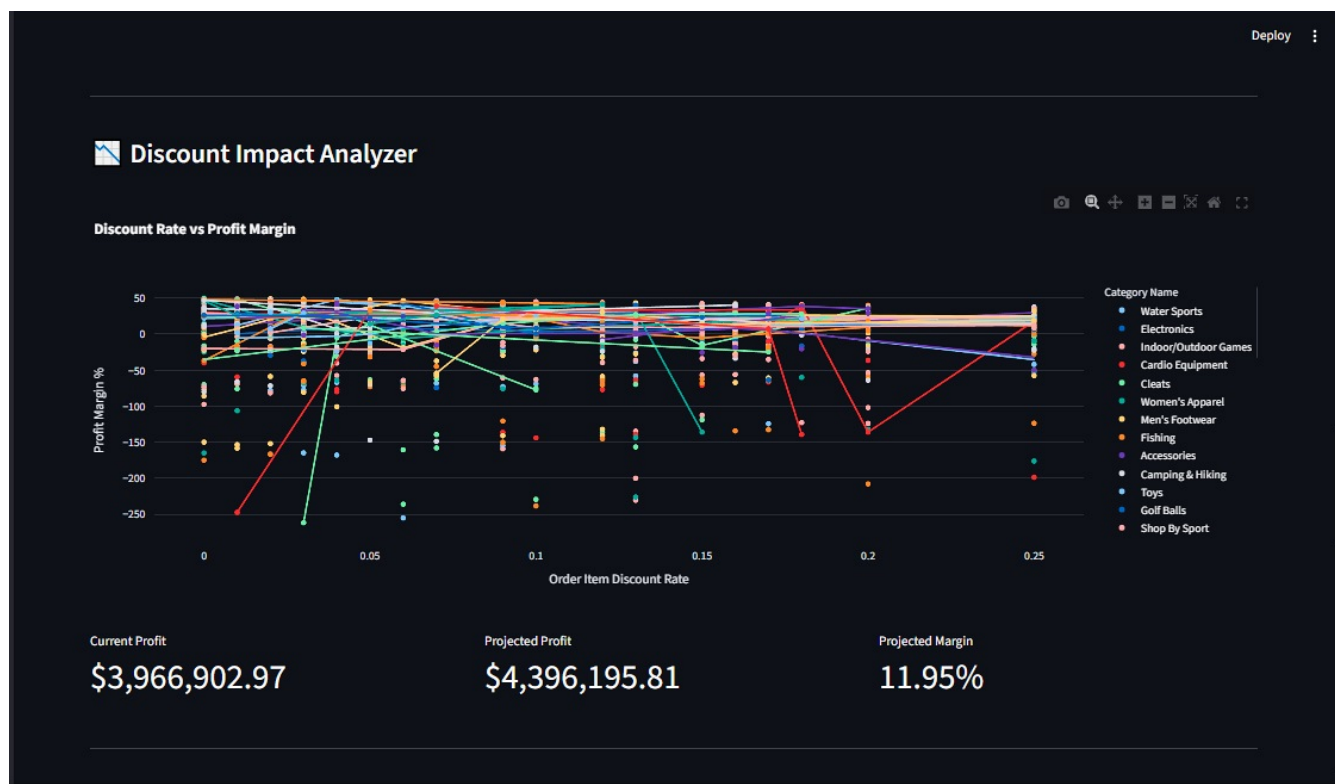


Figure 6. Discount impact analyzer showing discount rate vs profit margin and the result of a 15% discount cap simulation.

Using a 15% discount cap, the what-if simulation estimates that profit could increase from \$3,966,902.97 to \$4,396,195.81. The projected margin improves from 10.78% to 11.95%. This does not mean every discount above 15% should be removed immediately, but it does show that a clear approval rule could protect a significant amount of profit.

10. Market and Regional Profitability

The market view shows that USCA has the strongest margin at about 11.14%, followed closely by Africa and LATAM. Pacific Asia has the weakest market margin among the five markets shown, at about 10.37%. The differences are not extremely large, but in a high-volume logistics business even one percentage point can represent a meaningful amount of profit.



Figure 7. Market profitability chart comparing profit margin across Africa, Europe, LATAM, Pacific Asia, and USCA.

The regional chart and country map give another layer of detail. Western Europe and Central America are the strongest regions by total profit, each contributing more than \$600K. The country-level map helps stakeholders identify where commercial value is concentrated geographically.

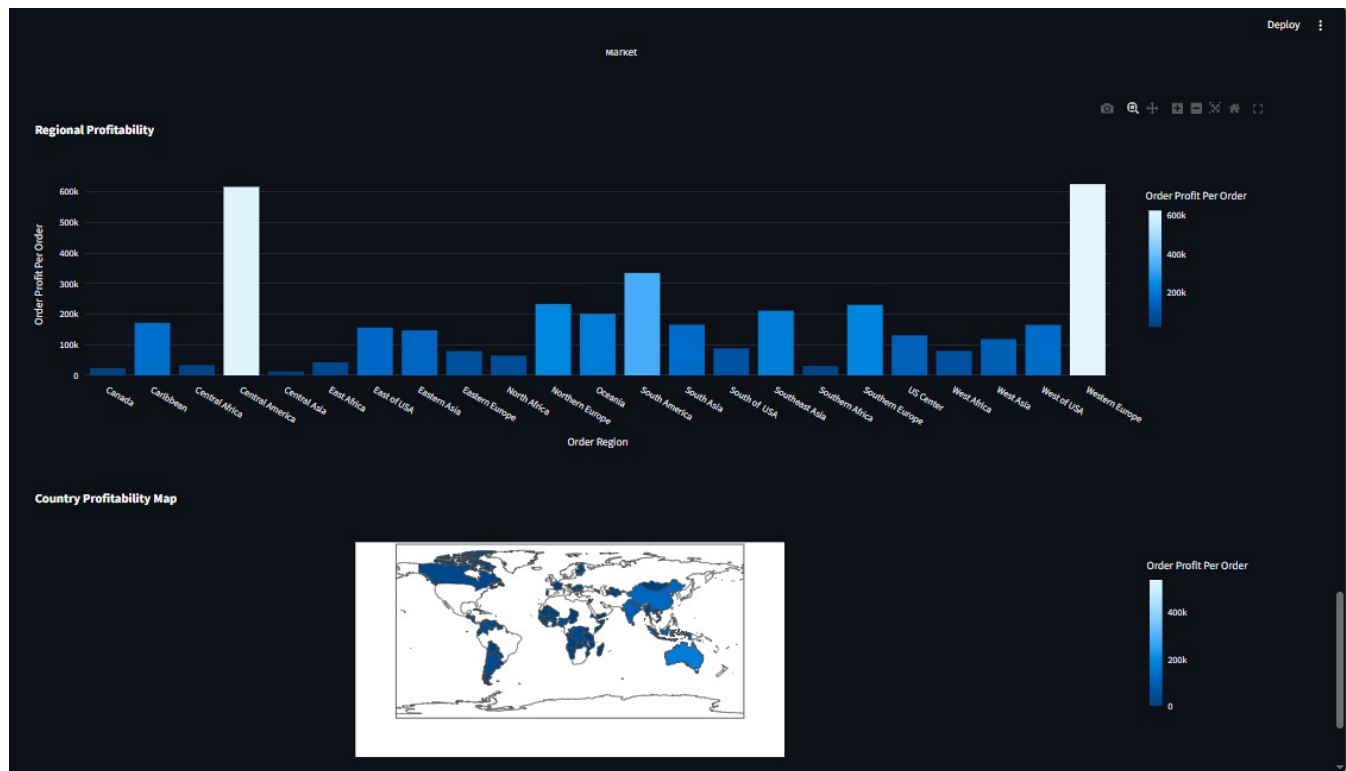


Figure 8. Regional profitability and country profitability map showing geographic profit distribution.

11. Key Findings

- The business generated \$36.78M in revenue and \$3.97M in profit, with a margin of 10.78%.
- Consumer customers produce the largest revenue share, but customer value needs to be measured using profit, not revenue alone.
- Customer profitability is concentrated. Strong customers generate more profit than the final net result, while loss-making customers reduce total profit.
- Fishing is the strongest absolute profit category, while Golf Bags & Carts has the best margin.
- Strength Training has the weakest margin and should be reviewed for pricing, shipping cost, or discount policy issues.
- A 15% discount cap scenario increases projected profit to \$4.40M and improves margin to 11.95%.
- USCA has the strongest market margin, while Western Europe and Central America are the strongest regions by total profit.

12. Recommendations

1. Build discount approval rules. Discounts above 15% should require review because the dashboard shows a clear opportunity to recover profit through better discount discipline.
2. Manage customers by profit tier, not only by sales volume. Platinum and Gold customers should receive stronger retention focus, while low-margin or loss-making accounts should be reviewed for pricing changes.

3. Review underperforming categories. Strength Training should be investigated first because its margin is far below the overall business margin.
4. Use category-region heatmaps in pricing reviews. A category can perform differently by region, so a single global price or discount strategy may not be enough.
5. Expand market intelligence reporting. Market and regional margin should be reviewed regularly so leadership can identify where growth is profitable and where it is only revenue growth.

13. Conclusion

The project shows why supply chain analytics should include commercial intelligence, not only operational efficiency. APL Logistics can deliver more value by understanding which customers, products, categories, and regions truly create profit. The Streamlit dashboard makes this practical by allowing decision makers to filter the data, test discount scenarios, and compare profitability across multiple business dimensions.

The main lesson from the analysis is that revenue is only the starting point. Profitability depends on customer mix, product mix, discount discipline, and regional execution. By acting on these insights, the business can protect margin and make better commercial decisions without losing sight of operational performance.

References and Tools Used

Dataset: APL Logistics cleaned supply chain transactions dataset provided for project analysis.

Tools: Python, pandas, NumPy, Streamlit, Plotly, and ReportLab.